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The impact of telework on emotional experience: When, and for whom, does telework improve daily affective well-being?

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Teleworking has become increasingly popular in organizations around the world. Despite this trend towards working outside of the traditional office setting, research has not yet examined how people feel (i.e., their affective experiences) on days when working at home versus in the office. Using a sample of 102 employees from a large US government agency, we employed a within-person design to test hypotheses about the relationship between teleworking and affective well-being. We also examined four individual differences (openness to experience, rumination, sensation seeking, and social connectedness outside of work) as cross-level moderators. Results show that employees experience more job-related positive affective well-being (PAWB) and less job-related negative affective well-being (NAWB) on days when they were teleworking compared to days they were working in the office. Findings show that several of the individual differences moderated the relationships. Discussion focuses on the need to consider the affective consequences of telework and the characteristics that determine who will benefit more or less from working at home.

Keywords: Telework; Personality; Negative affect; Positive affect.

Teleworking is now a common global practice. Approximately one-fourth (24%) of the Americans report working from home some hours each week (United States Bureau of Labor Statistics, 2011), and many European countries have similarly high teleworking rates (e.g., 24% in the Sweden, 29% in Finland; Flexibility.co.uk Limited, 2012). Although not uniformly positive, telework potentially can produce various individual and organizational benefits such as increased job satisfaction, lower work–family conflict, decreased turnover and absenteeism, and reduced costs for office space use and maintenance (Daniels, Lamond, & Standen, 2000; Gajendran & Harrision, 2007; Reason Foundation, 2005). Also, the view that telework is a cost-efficient alternative to the traditional work environment has led government agencies and other organizations to implement policies to allow and encourage telework (e.g., Telework Enhancement Act, 2010).

Although there has been increasingly widespread implementation of telework programs in both governmental and private sector organizations (Matevka, Rapino, Landivar, 2012), research has not kept pace with the questions raised by practice. Here, we address a set of seemingly significant, yet largely neglected, issues. First, we examine the emotional experience of working at home versus in the office. To date, relevant work on telework and well-being has primarily focused on attitudinal variables as the main well-being outcomes of interest (e.g., job satisfaction, Gajendran & Harrison, 2007; Vega, Anderson, & Kaplan, 2014). However, affective experiences on a given work day are clearly discrepant from one-time overall evaluations about one's job or circumstances. Indeed, Weiss (2002) describes how job satisfaction represents a broad evaluation of one's job (an attitude), whereas affect encompasses a variety of feelings (i.e., joy, anxiety) that may fluctuate during the work day. Affect and satisfaction are conceptually distinct, and thus, we extend the literature by considering the relationship between telework and affective experiences. To do so, we use a within-person methodology to best capture these fluctuations in affect.

Second, there is almost no work on individual differences, like personality traits, that may moderate the

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potential benefits of telework. Although research has considered contextual factors (e.g., supervisor support, technical support; Haines, St-Onge, & Archambault, 2002) that influence telework outcomes, researchers have not yet examined the relationship between employee traits and telework outcomes (for an exception, see O'Neill, Hambley, & Bercovich, 2014). As such, we do not know who would benefit more or less from working outside the traditional office environment. Exploring the impact of telework on emotional experience and who benefits most from telework can provide insight for employees as to whether teleworking would be beneficial as well as for organizations when implementing telework programs and determining which employees may be best suited to work outside of the office.

In this paper, we attempt to begin redressing these voids by providing insights into (1) the relationship between telework and affective well-being and (2) the individual differences that may moderate this relationship. To these ends, the remainder of the paper unfolds as follows. First, we present a theoretical rationale for the relationship between teleworking and positive and negative affective well-being (PAWB and NAWB). We then provide theoretical explanations for four individual difference variables predicted to moderate the telework–affective well-being relationships. Using a sample of employees from a large government agency, we test our hypotheses with a within-person design. We conclude with a synthesis of our findings as well as a discussion of the implications of our findings for research and telework in practice.

TELEWORK AND AFFECTIVE WELL-BEING

Given the significant proportion of workers who are now teleworking, it is essential to consider their affective well-being. Affective well-being has been linked to a variety of business outcomes including increased profitability, productivity, and lower rates of turnover (Harter, Schmidt, & Keyes, 2003) and positive individual outcomes such as job performance and physical health (Lyubomirsky, King, & Diener, 2005). As such, determining the extent to which work location influences employee affective well-being can provide insight into a potential mechanism (telework) through which organizations could impact employee well-being (and therefore business outcomes).

We focus on two aspects of job-related affective well-being—positive and negative affect. Higher positive affect is associated with states such as enthusiasm, alertness, and happiness, whereas higher negative affect includes negative feelings such as fear, anxiety, and guilt (Watson, Clark, McIntyre, & Hamaker, 1992). A large body of research suggests that positive and negative affect are two separate dimensions that are largely independent of each other (e.g., Burke, Brief, & George, 1993) and that occur through different biological and psychological mechanisms (Watson, 2000). In addition, positive and negative affect are largely associated with different antecedents and outcomes (e.g., Watson & Pennebaker, 1989). Given these findings, we examine positive and negative affect as two separate outcomes of telework.

We base our hypotheses on affective events theory (AET; Weiss & Cropanzano, 1996). AET proposes that experiencing different work events can impact employees' affective states (and subsequent attitudes and behaviours). AET focuses on the role of everyday happenings on the job as opposed to the stable features of the job and work context. Everyday events can include both minor (e.g., accomplishing a small task, everyday social interactions) and major events (e.g., receiving a promotion). Research has supported AET's propositions, indicating that experiences of positive or negative events are related to positive or negative moods, respectively (e.g., Miner, Glomb, & Hulin, 2005; Zohar, 1999).

Here, we apply and extend AET to consider the emotional impact of events occurring both at and beyond one's physical workplace (i.e., "the office"). We expect that, on average, employees will encounter different events when working at home versus in the office. Further, we expect that personality will impact the experience of these events—through differential exposure, perception, or reactions to them. Our model, based on AET, is presented in Figure 1. Next, we provide an overview of the rationale for why we expect that individuals working from home will experience more positive affect.

Positive affective well-being

According to AET, employees who experience positive events will experience positive emotions. There are several aspects of the telework environment that may lead to increased frequency of positive events and therefore result in more positive emotions. Researchers have established that teleworkers experience perceived autonomy (Gajendran & Harrision, 2007) because they have greater choice in the location and scheduling of their work tasks (e.g., DuBrin, 1991; Standen, Daniels, & Lamond, 1999). Teleworking has also been shown to be associated with higher feelings of control and flexibility (Huws, Korte, & Robinson, 1990; Maruyama & Tietze, 2012; Standen et al., 1999; Tremblay, 2003) which have been associated with mental health and well-being (Thompson & Prottas, 2006; Warr, 2007). Further, teleworkers experience less interruptions (Bailey & Kurland, 2002; Duxbury, Higgins, & Neufeld, 1998; Haddad, Lyons, & Chatterjee, 2009) which can lead to more goal progress, a commonly cited source of positive emotions (e.g., Brunstein, 1993). In sum, overall characteristics of the telework environment (increased autonomy, control, schedule flexibility, decreased interruptions, and increased ability to accomplish goals) suggest that teleworking should be

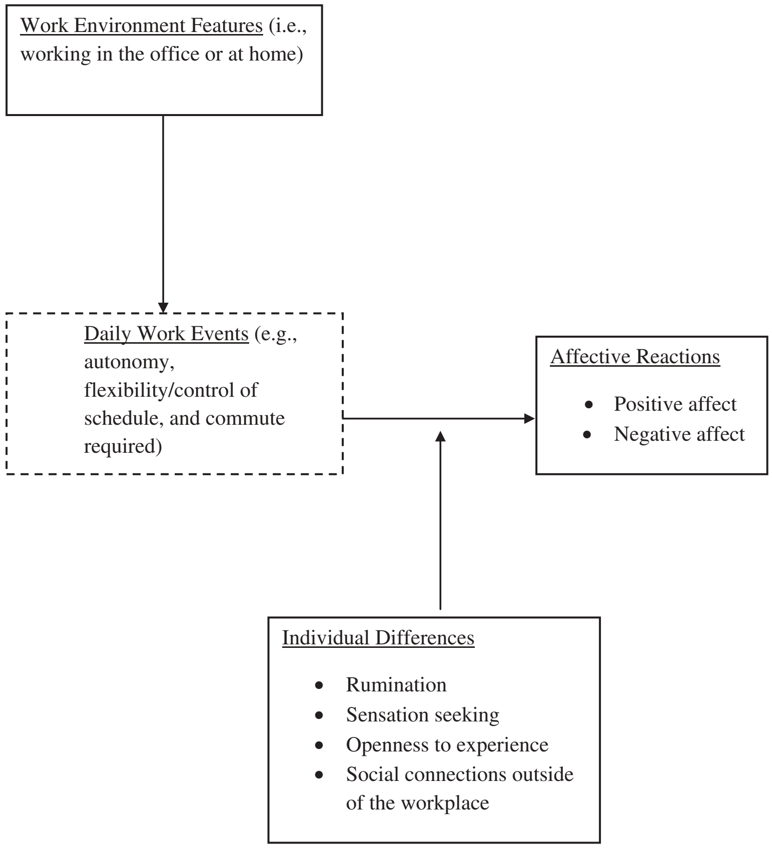

Figure 1. Our proposed model, based on affective events theory (Weiss & Cropanzano, 1996). Solid lines indicate that the constructs are measured in the current study and dashed lines indicate that the constructs are not measured in the current study.

associated with increased experiences of positive events.[1] Thus, we expect that:

> *Hypothesis 1:* When teleworking, employees will experience more job-related PAWB compared to when they are working at the office.

Negative affective well-being

In addition to the reasons why we would expect increased positive affect, we propose several additional reasons why we expect that employees will experience less negative affect while teleworking. AET indicates that employees will experience negative emotions after negative events. In general, survey research indicates that teleworkers experience less stress than office workers (World at Work, 2011). There are several reasons why working at home may result in a reduced number of negative events and less stress during the work day. First, working at home can lead to reduced interruptions (Bailey & Kurland, 2002; Duxbury et al., 1998; Haddad et al., 2009), a commonly cited source of stress and negative workplace affect (Jett & George, 2003). Second, telework reduces/eliminates commuting time (e.g., Peters, Tijdens, & Wetzels, 2004; Tremblay, 2003). Commuting is among the daily activities associated with the highest levels of negative affect (Kahneman, Krueger, Schkade, Schwarz, & Stone, 2004). Reduced commuting time could reduce negative affect in several ways such as by providing the opportunity for more sleep and/or exercise which are both associated with less negative emotions (Guszkowska, 2004; Motomura et al., 2013) and by reducing the amount of hassles and costs associated with transportation. In sum, we expect that the combination of reduced commuting time, less work-related interruptions, more autonomy, and more control will result in employees experiencing less negative affect when teleworking compared to working in the traditional office setting. Formally, we expect that:

> *Hypothesis 2:* When teleworking, employees will experience less job-related NAWB compared to when they are working at the office.

[1] Worth noting is that, instead of enhancing well-being, telework may, in some instances, decrease positive well-being. For example, research shows teleworkers perceive a reduction in visibility and decreased career development opportunities (Duxbury et al., 1998; Shamir & Salomon, 1985). However, here, we are focusing on affective experiences among individuals who telework regularly but not exclusively (a set amount of days per pay period), so we believe that these employees will experience more of the positive benefits of teleworking due to the fact that the negative aspects of telework tend to manifest their effects over longer periods of time among exclusive teleworkers (i.e., reduced face time resulting in stifled career advancement).

INDIVIDUAL DIFFERENCES AS MODERATORS OF THE TELEWORK–AFFECTIVE WELL-BEING RELATIONSHIPS

Although we expect telework to generally produce salubrious affective consequences, these effects may not be uniform. According to the AET framework, the emotional impact of work events varies as a function of theoretically relevant individual differences (see Weiss & Cropanzano, 1996). Supportive of this notion, substantial research documents that individual differences are key drivers of the ultimate affective outcomes resulting from workplace features and events (e.g., Judge, Erez, & Thoresen, 2000). These dispositional factors operate through several mechanisms, including influencing exposure to, perceptions of, and reactions to relevant events and contextual factors (Barsky, Thoresen, Warren, & Kaplan, 2004). Borrowing from these findings, we propose that the generally favourable effects of working from home will also vary as a function of key traits.

To date, the relevant literature is relatively mute on the potential role of individual differences in telework. Although a few studies have examined how individual differences such as personality traits relate to attitudes towards telework (Clark, Karau, & Michalisin, 2012; Gainey & Clenney, 2006), there is almost no research focusing on the relationship between these variables and outcomes of telework. To our knowledge, the only such study is a recent cross-sectional investigation of the relationships among several personality characteristics (conscientiousness, agreeableness, honesty, and neuroticism) and cyber slacking among employees who work remotely at least once a month (O'Neill et al., 2014). Here, we seek to expand the literature on this topic by broadening the scope of individual differences being researched, as well as considering how these individual differences affect the relationship between telework and affective well-being.

Although there are various potential characteristics that may moderate the effects of teleworking on affective well-being, consideration of the relationships proposed earlier led us to focus on four specific characteristics. We selected these traits based on theoretical considerations suggesting their role in impacting the day-to-day affective experience of working at home versus in the office. Based on the nature of events discussed earlier, we determined that resultant psychological states such as feelings of social disconnectedness, boredom, freedom and uncertainty about working in an environment with less structure, and increased time for self-reflective thoughts may largely differentiate the experience of working at home versus from the office. Working backwards, we then chose to focus on personal characteristics that should influence the amount and nature of these potential resultant reactions. The hypothesized impact of these four individual differences is discussed later.

Openness to experience

The first individual difference variable we chose to examine was openness to experience. Individuals who are high on openness to experience tend to be creative, broad-minded, curious, grasp new ideas quickly, and desire variety (Barrick & Mount, 1991; McCrae & Costa, 2003). Teleworking requires employees to adapt to a new work setting and structure, as well as to incorporate new technologies in the way that they perform their jobs (e.g., videoconferencing; Haines et al., 2002). A certain degree of openness to new experience seems beneficial in helping employees adapt to telework. In addition, the flexibility of telework in terms of scheduling work tasks and work hours should be particularly beneficial to individuals who are high in openness. Indeed, research has demonstrated that there is a positive relationship between openness and positive perceptions of telework programs (Gainey & Clenney, 2006), especially with the flexibility aspect of telework (Clark et al., 2012). Luse, McElroy, Townsend, and DeMarie (2013) found that openness to experience was the only one of the Big Five personality traits that significantly related to preference for virtual team work over face-to-face work. Given these reasons, we expect that openness to experience will moderate the relationship between teleworking and affective well-being. Specifically, we expect:

> *Hypothesis 3a:* Individual levels of openness to experience will moderate the relationship between teleworking and positive affect such that the relationship becomes more positive as openness increases.
>
> *Hypothesis 3b:* Individual levels of openness to experience will moderate the relationship between teleworking and negative affect such that the relationship becomes more negative as openness increases.

Rumination

A second potential moderating variable we examined was trait rumination. Rumination is a way of coping with negative emotions that involves repetitive and passive attention on one's negative emotion and the meaning of one's negative feelings (Nolen-Hoeksema, 1991, 2000; Nolen-Hoeksema, Larson, & Grayson, 1999). Although rumination can vary across situations, it is often conceptualized as an individual difference variable (Kuo et al., 2012; Treynor, Gonzalez, & Nolen-Hoeksema, 2003) that represents an individual's propensity to engage in ruminative thinking. Individuals who ruminate are often attempting to develop a better understanding of the cause of negative moods with the goal of ultimately feeling better. However, rumination instead actually tends to intensify negative mood by increasing attention to the negative mood (Lyubomirsky & Nolen-Hoeksema, 1995). Indeed, rumination has been linked to various negative outcomes such as increased severity of negative affect and depression (Nolen-Hoeksema, 1991).

We see two reasons why higher levels of rumination may attenuate the potential affective benefits of telework. First, rumination may exacerbate the effects of diminished social contact associated with telework (Golden, Veiga, & Dino, 2008). Research shows that employees tend to have more ruminative thoughts when they are alone compared to when they are around others (Cropley & Purvis, 2003). As such, those predisposed to ruminate may be at especially great risk to engage in such thinking while teleworking. Second, insofar as telework can represent a psychological "break" from some of the stressors of the workplace (see earlier), those lower in rumination may be more likely to benefit from that break. In contrast, those predisposed to telework may instead fail to reap the potential affective benefits of working at home, instead spending those days dwelling on work-related stress. We base this prediction on research showing that rumination is associated with work-related fatigue and an inability to psychologically detach from work-related problems (Querstret & Cropley, 2012). Given these two explanations, we offer the following hypotheses.

> *Hypothesis 4a:* Individual levels of rumination will moderate the relationship between teleworking and positive affect such that the relationship becomes more negative as rumination increases.
>
> *Hypothesis 4b:* Individual levels of rumination will moderate the relationship between teleworking and negative affect such that the relationship becomes less negative as rumination increases.

Sensation seeking

Third, we considered sensation seeking. Sensation seeking is considered a propensity to "seek novel and intense sensations and experiences" (Zuckerman & Como, 1983, p. 381). Individuals who are high in sensation seeking tend to need higher levels of stimulation to reach their optimal level of arousal and experience unpleasantness when they do not reach this level of arousal (Larsen & Buss, 2008). Research on sensation seeking in the management domain primarily has focused on occupational choice among sensation seekers (Roberti, 2004) and shown that sensation seekers tend to enjoy occupational environments that involve stimulating surroundings (Kish & Donnenwerth, 1969; Roberti, 2004). Research suggests that individuals working at the office experience greater physiological arousal (e.g., blood pressure) compared to teleworking individuals and also that the number and variety of social interactions are greater among workers at the office (Lundberg & Lindfors, 2002). As such, we would expect sensation seekers to enjoy working in such an environment more so than working from home where they have less access to such interaction.

> *Hypothesis 5a:* Individual levels of sensation seeking will moderate the relationship between teleworking and positive affect such that the relationship becomes more negative as sensation seeking increases.
>
> *Hypothesis 5b:* Individual levels of sensation seeking will moderate the relationship between teleworking and negative affect such that the relationship becomes less negative as sensation seeking increases.

Social connectedness

The last moderating variable we included was social connectedness outside of the workplace. Social connectedness outside of the workplace involves feeling in-touch and emotionally connected to individuals outside of one's job (Hawthorne, 2006). Social connectedness and social affiliation are strong predictors of well-being as they relate to meeting the basic psychological need of relatedness (Deci & Ryan, 1985; Diener, Suh, Lucas, & Smith, 1999). Previous studies have shown a positive relationship between social contact and affect at the between- and within-person level, and these studies indicate that maintaining social connectedness is key to maintaining well-being on a daily basis (Reis, Sheldon, Gable, Roscoe, & Ryan, 2000; Watson et al., 1992).

With respect to work location, research documents that working is associated with greater social interaction (e.g., more meetings and discussions) compared to teleworking (Lundberg & Lindfors, 2002). Working in a separate location from one's co-workers has the potential to cause feelings of isolation due to this lowered amount of interaction. However, if employees meet their need for connection through social interactions outside of work (e.g., through interactions with family, friends, volunteer, or community activities), then this may compensate or buffer against any feelings of disconnectedness that are experienced during the telework day. We expect that:

> *Hypothesis 6a:* Individual levels of social connectedness outside of work will moderate the relationship between teleworking and positive affect such that the relationship becomes more positive as social connectedness outside of work increases.
>
> *Hypothesis 6b:* Individual levels of social connectedness outside of work will moderate the relationship between teleworking and negative affect such that the relationship becomes more negative as social connectedness outside of work increases.

In sum, we expect that, overall, individuals will experience more PAWB and less NAWB while teleworking compared to while at the office due to increased experiences of positive and negative events. In addition, we predict that individuals who are higher in openness to experience and who have higher levels of social connectedness outside of the office will benefit most (experience higher positive affect and lower negative affect) while teleworking. We expect that employees who are higher in sensation seeking and rumination will not benefit relatively as much (experience lower positive affect and higher negative affect) from the telework environment. Next, we describe our within-person methodology used to test these hypotheses.

METHOD

Participants

We invited 702 employees from a large US federal agency to participate in the study. These individuals had previously signed telework agreements, indicating that they telework at least once per pay period. From this initial pool, 102 employees agreed to participate and provided data during at least one of the 4 days. These 102 individuals comprised the final sample. On average, respondents teleworked 2.88 days per week and had an average of approximately 3 years of experience teleworking (36.36 months). The following proportions of the respondents were teleworking on each day of the survey: 59.80% on day 1, 69.23% on day 2, 71.76% on day 3, and 72.97% on day 4. Respondents were 50% female, and the sample represented a range of age groups, with 3.65% less than 25, 25.61% between 26 and 35, 19.51% between 36 and 45, 28.05% between 46 and 55, and 20.73% between 56 and 65 years old.[2] The participants' exact job titles varied but generally involved business operations management and contract support.

Procedure

Data were collected from a sample of employees from a large US government organization. Participants who usually telework at least 1 day per pay period were invited to participate in the survey at four time points over a period of 2 weeks. Surveys were sent on Monday and Wednesday of week 1 and Tuesday and Thursday of week 2. Given that employees at the organization have varied teleworking schedules, we collected data on four different days of the week in order to capture the days of the week when individuals would likely be teleworking. Many individuals from the organization do not work on Fridays due to compressed work schedules; hence, this day was not included.

Participants were asked to complete the survey each of the 4 days and to make at least one of those four responses on a day they were teleworking. On those 4 days, participants received an e-mail that contained a link to the secure survey website at 2:00 p.m. and were asked to complete the measure before the end of the workday.

In order to track respondents across the four surveys/days while still protecting the anonymity of the employees, a series of three questions were asked in order to create a unique identifier for each respondent. These three items asked respondents to report their high school mascot, the name of their first pet, and their shoe size. The three items were presented at the beginning of each of the four surveys. In this way, the same individual's responses can be tracked and compared across days.

Measures

Telework status

At the beginning of each survey, participants reported whether they were working from home or at the office that day.

Job-related affective well-being

At each time point, participants were asked to complete 10 items from the Job-Related Affective Well-Being Scale (JAWS; Van Katwyk, Fox, Spector, & Kelloway, 2000). This is a widely used scale of job-related affect (see http://shell.cas.usf.edu/~pspector/scales/jawspage.html). Five items measured PAWB (at ease, grateful, enthusiastic, happy, and proud) and five items measured NAWB (bored, frustrated, angry, anxious, and fatigued). Due to practical constraints, we were only able to include 10 items from the full 30 item JAWS, and we selected the items that we expected would be most commonly experienced at work (e.g., we did not include "disgusted," "frightened," or "ecstatic"). We also included a mixture of "high" and "low" arousal items from each of the positive and negative scales. The stem question was rephrased to put emphasis on the particular day the survey was taken (as opposed to the original JAWS which references the past 30 days). The stem read, "Below are a number of statements that describe different emotions that a job can make a person feel. Please indicate the amount to which any part of your job (e.g., the work, coworkers, supervisor, clients, pay) has made you feel that emotion TODAY." The items read "My job made me feel…." The response scale ranged from strongly disagree (1) to strongly agree (5). Exploratory factor analyses at each of the four time points support a two-factor solution, and

[2]Given that the non-respondents did not provide any data, we were not able to compare the respondents versus non-respondents on demographic or study variables. This said, we could not think of factors that would have led the current sample to be non-representative of the teleworkers in this agency. In addition, the sample demonstrated variance on key variables such as telework frequency, age, gender, supervisor status, and tenure.

TABLE 1
Observation level correlation matrix

	Mean	SD	*1*	*2*	*3*	*4*	*5*	*6*
1. Telework status	.67	.47						
2. Positive affect	3.74	.81	−.07					
3. Negative affect	2.33	.95	−.04	−.49**				
4. Sensation seeking	2.88	.57	−.01	−.21**	.07			
5. Rumination	2.07	.58	.02	−.11	.18*	.11		
6. Openness	3.38	.29	.01	.06	−.13	−.05	−.01	
7. Social connectedness outside of the workplace	3.35	.40	.18	−.12	.06	.00	.36**	.08

Telework status was coded 0 and 1, where 1 indicates the employee was teleworking on that day. Positive and negative affect represent means across the four time points.
$^*p < .05$, $^{**}p < .01$.

across the four time points, the internal consistency reliability (α) for PAWB ranged from .87 to .93 and for NAWB from .79 to .88.

Individual difference measures

During the first survey administration, participants were asked to complete measures of openness to experience, sensation seeking, trait rumination, and social connectedness outside of the workplace. Participants were also asked to respond to several demographic items (gender, experience teleworking, number of days teleworked per week on average, and age).

Openness to experience. The measure of openness to experience included 10 items (Goldberg, 1999). Participants were asked to report the extent to which they agreed or disagreed with the statements on a 1–5 scale (strongly disagree to strongly agree). The scale had sufficient internal consistency reliability (α = .83). A sample item is "I have a vivid imagination."

Trait rumination. The trait rumination scale was adapted from Treynor et al. (2003) and included five items. The question stem read, "We all feel sad, blue, and down sometimes. People think and do many different things when feeling this way. Please indicate how frequently you do each of the following when feeling sad or down. Please indicate what you generally do, not what you think you should do." The response scale ranged from 1 to 5, where 1 = never and 5 = always. The scale had satisfactory internal consistency reliability (α = .75).

Sensation seeking. A five-item scale was adapted from an existing survey (Zuckerman, Kolin, Price, & Zoob, 1964) to measure sensation seeking. Participants were asked to report the extent to which they agree or disagree on a 1–5 scale. Initial internal consistency reliability was α = .48. After observing that the reliability was so low, we conducted an exploratory factor analysis (using Principle Axis Factoring and Promax rotation) to determine whether there were multiple dimensions. We found that one of the five items ("I find certain pleasure in routine kinds of work") did not load on the same factor as the other four items. This item was removed from the scale, and the resulting internal consistency reliability increased to α = .66.

Social connectedness outside of the workplace. The measure of social connectedness outside of the workplace included a stem question with five items measured on a 1–5 scale (strongly disagree to strongly agree) (adapted from Hawthorne, 2006). The stem read, "Please answer the following questions about yourself outside of the workplace. During the past four weeks…" and a sample item is, "It has been easy to relate to others." The scale had adequate internal consistency reliability (α = .73).

The mean, standard deviations, and an observation-level correlation matrix for these variables are provided in Table 1. The observation-level means indicate that respondents generally had fairly high levels of positive affect (M = 3.74, SD = .81) and low negative affect (M = 2.33, SD = .95) when averaging across the 4 days. Positive and negative affect were moderately negatively correlated at the observation level (r = −.49, p < .01). Only two of the individual difference variables were significantly correlated (social connectedness and rumination, r = .36, p < .01). Employees had average levels of sensation seeking (M = 2.88, SD = .57), openness to experience (M = 3.38, SD = .29), and social connectedness outside of the workplace (M = 3.35, SD = .40), and somewhat low levels of trait rumination (M = 2.07, SD = .58). Please refer to the Appendix for the full list of items in the measures.

RESULTS

Main effects

We used random coefficient modeling (RCM) to analyse the data since the observations are nested (time points nested within person). We employed Full Information Maximum Likelihood (FIML) estimation as our missing data treatment for all analyses. Our first two hypotheses focused on the main effect of telework on job-related PAWB and NAWB. Hypothesis 1 predicted that employees would experience more PAWB while teleworking compared to when working at the office. The telework variable was coded as 1 for teleworking and 0 for working at the office. The analysis shows a positive coefficient for telework in predicting positive affect which supports Hypothesis 1 ($\gamma = .15$, $p < .05$). We also hypothesized that employees would experience less negative affect while teleworking (Hypothesis 2), and this was also supported by the analysis as indicated by a significant negative coefficient ($\gamma = -.23$, $p < .05$). Thus, employees reported experiencing greater positive affect and less negative affect when working at home versus in the office.

Cross-level moderation effects

We assessed the four moderators by entering them as level-2 variables in the main effects models. The results from these analyses appear in Table 2. Hypotheses 3a and 3b predicted that openness would moderate the relationship between teleworking and positive affect such that the relationship becomes more positive as openness increased. The results show support for Hypothesis 3a ($\gamma = .74$, $p < .05$), indicating that openness is a cross-level moderator of the telework and positive affect relationship; the relationship becomes more positive as openness to experience increases. In contrast, we did not find support for Hypothesis 3b, indicating that openness does not influence the telework–negative affect relationship.

Next, we tested rumination as a cross-level moderator. Results support Hypotheses 4a, showing that the telework–positive affect relationship becomes more negative as trait rumination increases ($\gamma = -.38$, $p < .01$). However, the results do not support Hypothesis 4b; rumination does not moderate the telework–negative affect relationship. Turning to sensation seeking, we expected that, as sensation seeking increases, both the telework–positive affect and the telework–negative affect relationships would become more negative. However, our analyses did not support either of these hypotheses (Hypotheses 5a and 5b). Finally, we tested our hypotheses about individuals' social connectedness outside of the workplace. We found support for Hypothesis 6a, revealing that the relationship between telework and positive affect is moderated by one's social connectedness outside of the workplace such that the relationship becomes more positive as social connectedness increases ($\gamma = .75$, $p < .001$). The results also support Hypothesis 6b, showing that social connectedness outside of work moderates the relationship between telework and negative affect such that the relationship becomes more negative as social connectedness increases (individuals experience less negative affect while teleworking as social connectedness increases; $\gamma = -.73$, $p < .01$).

In sum, the results regarding the individual differences revealed that the relationship between telework and positive affect was more strongly positive for individuals higher in openness to experience, lower in trait rumination, and with greater social connectedness. Those with higher levels of social connectedness had a more strongly negative relationship between telework and NAWB. The other cross-level moderation effects were not significant. The significant interactions are plotted in Figures 2–5.

DISCUSSION

The purpose of this research was to explore whether work location influences employees' emotional well-being and if it does, for whom? In the following paragraphs we summarize the relevant study findings and discuss implications and potential extensions of these findings. We then note some limitations of the study, mention additional directions for future scholarly work, and describe the practical implications.

The first major topic we sought to address was whether working at home, versus in the office, influences day-to-day affective experience. The results revealed that working at home was generally associated with both greater positive affect and lower negative affect, in support of our hypotheses based on AET. We see these findings as having two major implications. A first

TABLE 2
Random coefficient modeling results

	Positive affect			*Negative affect*		
	γ	SE	t-*Value*	γ	SE	t-*Value*
Telework	.15*	.06	$t(173) = 2.30$	−.23*	.09	$t(173) = -2.49$
Telework × Rumination	−.38**	.13	$t(66) = -2.84$	.37	.19	$t(66) = 2.00$
Telework × SenSeek	−.12	.12	$t(66) = -.95$	.00	.22	$t(66) = .01$
Telework × Openness	.74*	.31	$t(66) = 2.43$	−.06	.49	$t(66) = -.13$
Telework × SocConn	.75**	.21	$t(66) = 3.54$	−.73**	.27	$t(66) = -2.70$

The main effects of individual differences on positive and negative affect were included in the analyses. We did not hypothesize any main effects and all main effects were non-significant ($p > .10$) with the exception of the main effect of social connectedness on positive affect ($\gamma = -.75$, $SE = .28$, $t(66) = 2.68$, $p < .001$).

*$p < .05$, **$p < .01$.

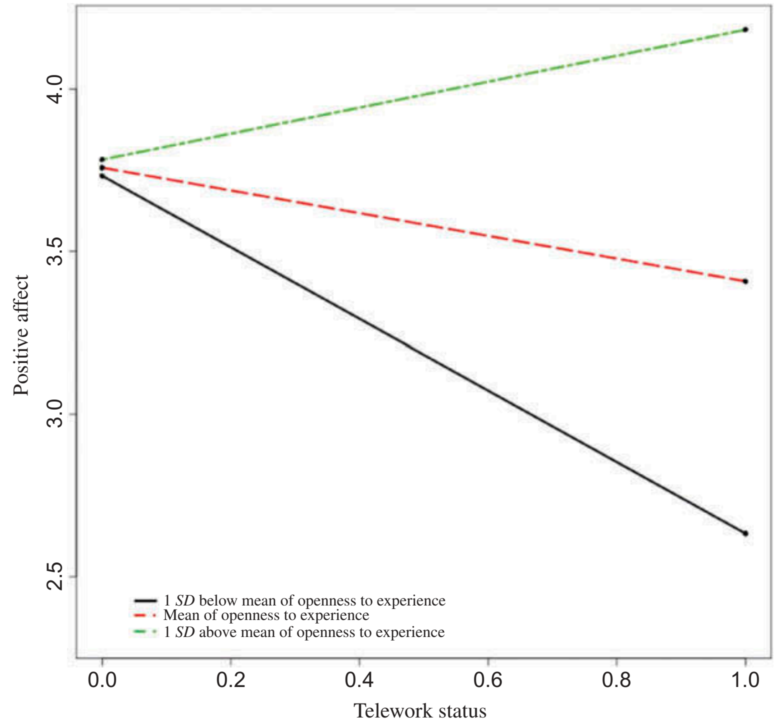

Figure 2. Cross-level interaction of openness and telework predicting positive affect. Plotted at one standard deviation below the mean, the mean, and one standard deviation above the mean of openness.

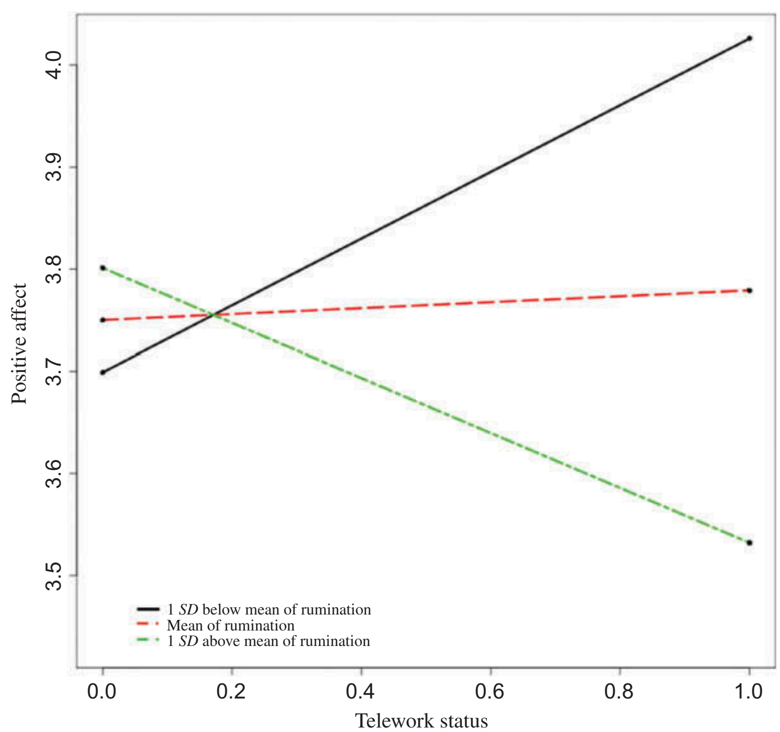

Figure 3. Cross-level interaction of rumination and telework predicting positive affect. Plotted at one standard deviation below the mean, the mean, and one standard deviation above the mean of rumination.

notable implication of these results is that the location one works on a given day actually impacts one's phenomenological emotional experience. To date, the dominant paradigm in the telework literature entails assessing attitudinal and/or performance outcomes, along with proposed mediators, either once (comparing

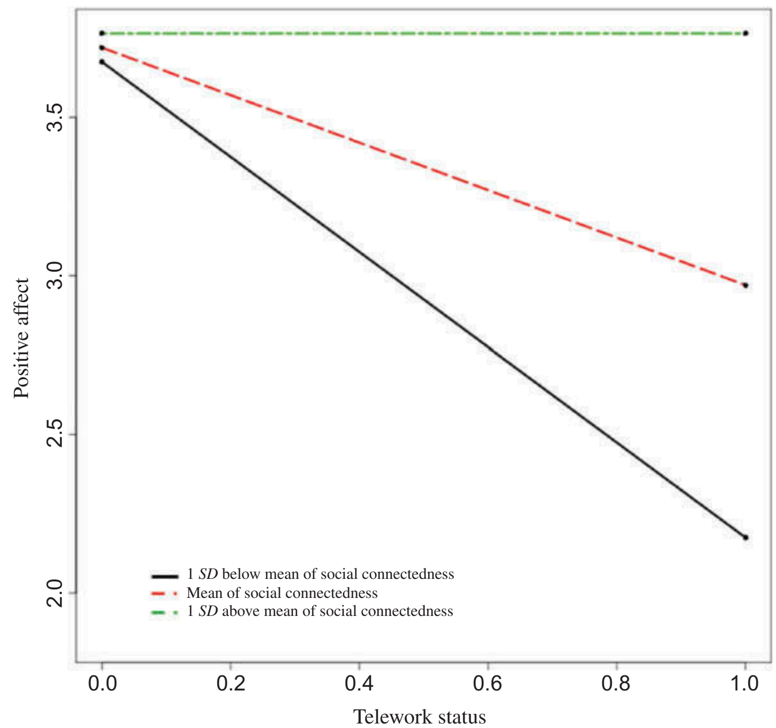

Figure 4. Cross-level interaction of social connectedness outside of the workplace and telework predicting positive affect. Plotted at one standard deviation below the mean, the mean, and one standard deviation above the mean of social connectedness.

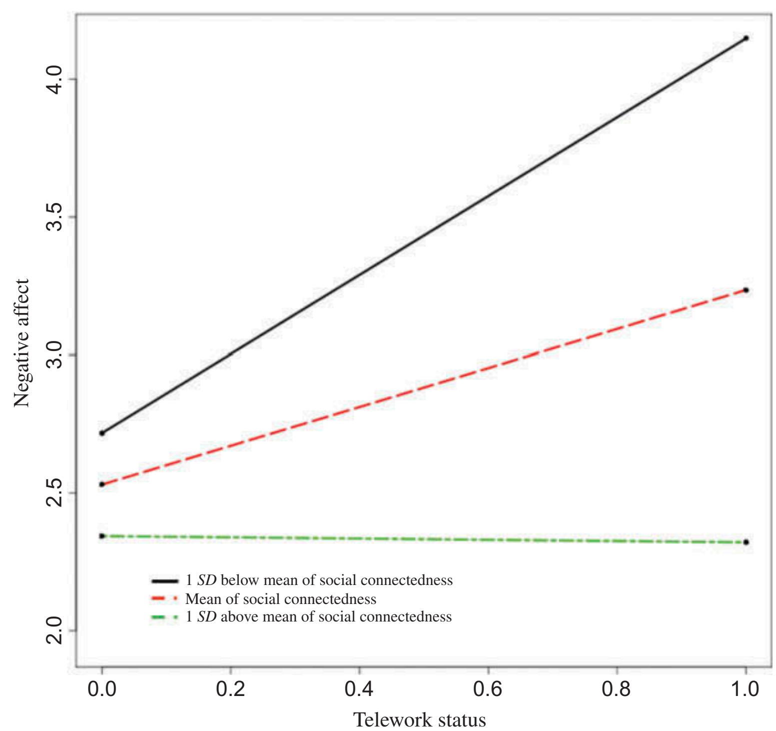

Figure 5. Cross-level interaction of social connectedness outside of the workplace and telework predicting negative affect. Plotted at one standard deviation below the mean, the mean, and one standard deviation above the mean of social connectedness.

teleworkers to non-teleworkers) or at multiple points, separated by several months (see Gajendran & Harrision, 2007). The current results show that working at home does not just lead to different broad evaluative judgments but that it actually *feels* differently from working in the office.

A second point regarding the affect findings is that telework impacted both negative and positive affect—decreasing and increasing them, respectively. To the extent that the telework literature has incorporated affect—in the form of role stress and work–family conflict—the focus has been on negative forms of emotion. In addition to more explicitly documenting that telework can reduce the experience of emotions like stress and anxiety, the present results reveal that it can also increase positive emotions like happiness, joy, and the like.

Of importance, though, the effect sizes for work location were actually rather modest. Much more interesting and significant in our view than these main effects are the current findings documenting that the affective consequences of telework seem to vary dramatically as a function of individual differences. Surprisingly, despite the intuitive notion that flexible work arrangements may be differentially beneficial for certain types of individuals, there is almost no research on personality and virtual work (see Luse et al., 2013 for an exception). Collectively, our results indeed demonstrate that several factors moderate the affective consequences of telework; it seems that the positive emotional effects of telework are not unconditional.

Turning first to rumination, the findings revealed that the typical boost in positive affect that telework can generate appears to be less strong for those who tend to ruminate. This result supported our hypothesis, suggesting that physically working alone may be associated with increased ruminative thoughts which lead to less positive emotion. However, our findings are discrepant with previous research on rumination and negative affect, which suggest that working alone may also lead to experiencing greater negative emotion (Cropley & Purvis, 2003).

Although we can only conjecture about this latter finding, one possibility has to do with the type of rumination under investigation here. Much of the research on rumination focus on thoughts about stressful, and oftentimes traumatic, events and circumstances (e.g., Ehring, Frank, & Ehlers, 2008). Given that most employees in the current sample likely were not experiencing traumatic scenarios, the content of ruminative thought while working generally may not have reflected or induced extreme levels of negative affect. Instead, given the lack of social contact and stimulation in this environment, teleworkers instead may ruminate about things like their (lack of) social relationships, their family circumstances, and/or the types of existential considerations associated with a lack of arousal (Barbalet, 1999). Those types of ruminations would seem more closely aligned with a lack of positive affect rather than with higher negative affect such as anxiety, anger, etc. (Watson, 2000). This said, future research should consider measuring the amount and content of ruminative thoughts that individuals experience while working in different environments to directly capture the underlying theoretical mechanisms.

In terms of openness to experience, we found that the telework–positive affect relationship becomes more positive as openness increases. Notably, in one of the few existing relevant studies on individual differences and alternative work arrangements, Luse and colleagues (2013) found that openness was the Big Five factor most strongly associated with a preference for virtual, versus face-to-face, teamwork. Future studies might try to dig deeper into the causes of this relationship. In particular, investigations may address whether those higher in openness prefer virtual working arrangements due to the flexibility such arrangements might provide (e.g., in constructing one's workspace at home in creative ways) and/or to the variability they provide (i.e., sometimes working face-to-face and other times in a virtual context).

In contrast to the relationship involving positive affect, our hypothesis about openness moderating the relationship between telework and negative affect was not supported. Thus, openness appears to boost positive affect among teleworkers but does not decrease negative affect. We can only conjecture on this finding, but one possibility is that the greater leeway and room for expression that telework provides (Gajendran & Harrision, 2007) is more closely linked to positive affect than it is to feelings like anxiety, guilt, and other aspects of negative affect.

Our findings regarding social connectedness outside the work place indicate that being engaged in social relationships outside of work can further enhance the relationship between telework and well-being. Individuals who were highly connected outside of work experienced even higher levels of positive affect and lower levels of negative affect. These findings support our hypotheses that meeting one's need for social affiliation outside of work can be beneficial among individuals who telework. Going forward, we would also suggest researchers examine this relationship among employees who work exclusively, or almost exclusively, from home. Plausibly, the current finding could either be stronger among those extreme teleworkers, given their greater isolation (Golden et al., 2008) or weaker, due to their potentially categorically different lifestyle (e.g., restructuring their time and social lives differently than partial teleworkers).

Finally, we did not find support for our hypotheses regarding the moderating effects of sensation seeking. One potential explanation for these findings is that we may have had a restricted range for the sensation seeking variable. Studies show that individuals who are high in sensation seeking tend to select highly stimulating professions such as firefighting, race car driving, and mountain climbing (Zaleski, 1984). Our sample consisted of federal government employees who worked primarily on finance and acquisition related tasks; thus, it is likely that employees in our sample did not represent the full range of sensation seeking. Another explanation could be that this variable does not moderate the relationship between telework and well-being because individuals who are high in sensation seeking may seek out excitement regardless of the location and thus their affect is not impacted by location.

In closing this section, we emphasize that that the present results suggest that different individual differences matter for different outcomes (positive or negative affect here) and, likely through impacting perceptions of, and reactions to, different types of events. Work location is obviously not an event per se, but it instead represents a context making the experience of different types of events, and resultant reactions, more or less likely. Here, we used these events and their potential reactions as the "jumping off point" in identifying theoretically plausible individual differences. The diversity of events (e.g., being bored versus feeling socially disconnected) led to an arguably diverse set of individual differences. As such, the present analysis and results did not derive from, and do not speak directly to, a given theoretical framework of individual differences.

This all said, these findings do provide some tentative clues as to what kind of framework might be useful to guide future similar investigations. Specifically, these four traits and the set of findings point to the possibility that working from home is associated with less externally generated stimulation and that this difference is a key driver of the current results. Whether deriving from less social interaction, less ambient noise, or other contextual factors, working from home is, or at least can be, a less stimulating than working in most offices. Research indicates that having less external stimulation can either decrease or increase internal stimulation (see Loukidou, Loan-Clarke, & Daniels, 2009). Very possibly, the effect depends on one's internal level of arousal, as those with greater pre-existing internal arousal need, and desire, less external stimulation than those with lower internal levels (e.g., Furnham & Allass, 1999; Larsen & Buss, 2008). Seen in this light, the present findings tentatively suggest that arousal, in some form (e.g., cognitive, physical, physiological), and optimal levels of it, is the key determinant of whether, or to what degree, working at home is emotionally beneficial. As such, we would also call for investigations of a trait like boredom proneness and for experience sampling studies incorporating measures of real-time arousal in different working contexts.

Limitations and future directions

Our study was not without some limitations. One limitation was that the sample only included individuals from a single, governmental organization. Upper management in this organization held a very favourable view of teleworking and, accordingly, instituted a liberal teleworking policy. Also, we were not able to compare participants and non-respondents on key variables. Given these considerations, we cannot be completely certain that the current findings would generalize to other employees or to employees in other, less supportive organizations. Regarding the latter consideration, employees who choose to telework in organizations where telework is not viewed as favourably may not enjoy the same benefits in emotional well-being when working from home as did the current sample.

More generally, the motives and consequences of telework may certainly vary significantly across organizations, due to varying organizational cultures and attitudes towards telework. Thus, our results provide initial insight into the phenomenon studied, but these results may or may not generalize to other organizations. Future studies should attempt to corroborate the present findings in other organizations as well as across other types of occupations. In addition, our sample was composed of employees who teleworked a set number of times per pay period. Future research exploring the generalizability of these findings to samples containing teleworkers with a variety of arrangements (e.g., occasional teleworkers, exclusive teleworkers, those who are forced to telework certain days) would be informative.

Another limitation of our study was that employees were not randomly assigned to telework, which limits our ability to draw inferences of causality. Given the practical constraints and the agency's existing telework policy, it was not possible to randomly assign employees to conditions. Although our within-person design provides more convincing evidence than cross-sectional designs, we recognize that several factors, such as personality or job attitudes, may have impacted self-selection into telework arrangements (a challenge also faced by other researchers; Bockerman, Bryson, & Ilmakunnas, 2012). Thus, our sample may have a restricted range on some of these characteristics. Worth noting is that such a restriction in range would have made it more difficult to find significant results. So, arguably, this limitation could actually suggest that the current results underestimate the true effects of telework on affect. This said, without knowledge regarding which individual differences impacted selection into telework programs, we cannot definitively know whether or not these impacted our estimates or in what direction. Future researchers may seek to randomly assign respondents to telework or no telework conditions to remove any variability due to personality or work attitude differences. In addition, future studies should measure work attitudes to determine the relationship between attitudes, affect, and physical work location.

Also, as alluded to above, future studies should consider mediating mechanisms through which telework influences affect. As proposed by the AET, work environments may impact employee affect via work events. We were not able to measure work events in the present investigation, but research would benefit from studies measuring specific events that employees experience on days when they are working in the office versus day when they are teleworking (e.g., a diary study). In addition, although future research could examine established between-person mediators between telework and attitudinal outcomes (e.g., autonomy; Gajendran & Harrision, 2007), there may be other, more novel, potential mediators at the within-person level. For instance, working at home may allow for greater physical activity or working while listening to music, factors associated with affective outcomes (e.g., Berger & Motl, 2000). In addition, telework may increase positive affect by fostering perceptions of competence. Studies have shown that teleworkers report higher levels of productivity compared to employees who work in the office (e.g., Gajendran & Harrision, 2007; Hill, Ferris, & Martinson, 2003; Hill, Miller, Weiner, & Colihan, 1998), and fulfilling the psychological need of competence (Deci & Ryan, 1991) may increase positive affect. Finally, the mediator of commuting time could be an area for future research. As mentioned in the introduction, commuting-related hassles are one of the largest contributors to negative affect (Kahneman et al., 2004) and thus could certainly be a mechanism through which telework reduces negative emotions. Ideas such as these seem ripe for investigation.

Finally, studies investigating other individual differences and the effects of individual differences on other outcomes would seem very useful. The current findings suggest that, at least in terms of affective gains, telework is more beneficial for some employees than others. We chose traits based on theoretical considerations but, obviously, no study can capture all potentially relevant characteristics. Intuitively, traits like boredom proneness (as noted earlier) and conscientiousness would also seem like candidates moderating the effects of working at home. Related to this notion, studies addressing the role of traits with respect to other outcomes also seem necessary. From an organizational perspective, outcomes like affective well-being presumably are not paramount; consequences like performance and productivity are. Notably, reviews of the relevant literature suggest that telework can, but does not always, lead to gains in performance and productivity (de Menezes & Kelliher, 2011; Gajendran & Harrision, 2007). Although various contextual factors likely moderate this effect (e.g., availability of rich media for communication, supervisory support for telework), individual differences also likely play a significant role and very well may interact with those situational features.

CONCLUSIONS AND PRACTICAL IMPLICATIONS

The present findings are the first of which we are aware to document the fluctuations in affect that result from working at home versus in the office. Also, they represent an initial exploration into the role of individual differences as moderators of those effects. The importance of individual differences in adapting to various work environments has been suggested by previous research, but we have extended these findings to the telework environment.

With respect to practical implications, these results would seem to have relevance for the implementation of telework policies and the management of teleworkers. Our findings suggest that allowing employees to telework can act as a simple intervention boost well-being among some employees. Managers and organizations must consider the individual differences of employees before assuming that telework will be beneficial for all. In general, managers should encourage teleworking employees to develop and maintain social connections outside of the workplace in order to increase well-being and to buffer against negative affect. In addition, the telework experiences of employees who are high in trait rumination and low in openness should be monitored by managers to determine whether telework is a worthwhile arrangement for them. In addition, organizations with telework programs may consider administering personality assessments to be used for employee self-awareness or monitoring of current teleworking employees. These personality assessments could also be used by career planning or development professionals to help clients determine whether jobs that involve teleworking would suit their personality. Taken together, our findings suggest that the relationship between work environment, individual differences, and well-being is complex and warrants careful consideration by organizational executives and scholars.

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APPENDIX
MEASURES

Affective well-being

Below are a number of statements that describe different emotions that a job can make a person feel. Please indicate the amount to which any part of your job (e.g., the work, co-workers, supervisor, clients, and pay) has made you feel that emotion TODAY.

Positive affective well-being

(1) My job made me feel at ease.
(2) My job made me feel grateful.
(3) My job made me feel enthusiastic.
(4) My job made me feel happy.
(5) My job made me feel proud.

Negative affective well-being

(1) My job made me feel frustrated.
(2) My job made me feel angry.
(3) My job made me feel anxious.
(4) My job made me feel fatigued.
(5) My job made me feel bored.

Openness

Please indicate the extent to which you agree or disagree with the following statements (strongly disagree to strongly agree).

(1) I have a rich vocabulary.
(2) I have a vivid imagination.
(3) I have excellent ideas.
(4) I am quick to understand things.
(5) I use difficult words.
(6) I spend time reflecting on things.
(7) I am full of ideas.
(8) I have difficulty understanding abstract ideas. (−)
(9) I am not interested in abstract ideas. (−)
(10) I do not have a good imagination. (−)

Social connectedness outside of the workplace

Please answer the following questions about yourself outside of the workplace. During the past 4 weeks… (strongly disagree to strongly agree).

(1) It has been easy to relate to others.
(2) I felt isolated from other people. (−)
(3) I had someone to share my feelings with.
(4) I found it easy to get in touch with others when I needed to.
(5) When with other people, I felt separate from them. (−)

Rumination

Please rate the extent to which you agree or disagree with the following statements (strongly disagree to strongly agree).

(1) I think “Why do I have problems other people don’t have?”
(2) I think about how I don’t feel up to anything.
(3) I think “What am I doing to deserve this?”
(4) I think about a recent situation, wishing it had gone better.
(5) I analyse my personality and try to understand why I am depressed.

Sensation seeking

Please rate the extent to which you agree or disagree with the following statements (strongly disagree to strongly agree).

(1) I get bored sometimes seeing the same old faces.
(2) I sometimes like to drive very fast because I find it exciting.
(3) I prefer friends who are excitingly unpredictable.
(4) I like doing things just for the thrill of it.